Basic Fourier Series Pairs

Time Domain $x(t) = \sum_{k=-\infty}^{\infty} X[k] e^{jk\Omega_o t}$ Period T	Frequency Domain $X[k] = \frac{1}{T} \int_{<T>} x(t) e^{-jk\Omega_o t} dt$ $\Omega_o = \frac{2\pi}{T}$
$x(t) = \begin{cases} 1, & \lvert t \rvert \le T_o \\ 0, & T_o < \lvert t \rvert \le T/2 \end{cases}$	$X[k] = \frac{\sin(k\Omega_o T_o)}{k\pi}$
$x(t) = e^{jp\Omega_o t}$	$X[k] = \delta[k-p]$
$x(t) = \cos(p\Omega_o t)$	$X[k] = \frac{1}{2}\delta[k-p] + \frac{1}{2}\delta[k+p]$
$x(t) = \sin(p\Omega_o t)$	$X[k] = \frac{1}{2j}\delta[k-p] - \frac{1}{2j}\delta[k+p]$
$x(t) = \sum_{p=-\infty}^{\infty} \delta(t - pT)$	$X[k] = \frac{1}{T}$